

Diadenosine Tetraphosphate (Ap4A)

Turnover: A Review-Style Synthesis

1. Executive Summary

Diadenosine 5',5'''-P¹,P⁴-tetraphosphate (**Ap4A**) is a dinucleoside polyphosphate built from two adenosine moieties bridged by four phosphates. It accumulates when cells are stressed and has been proposed, for more than four decades, to act as an intracellular "alarmone" or second messenger. This review treats Ap4A **turnover** as a self-contained, reusable metabolic module with exactly two operations — **synthesis** and **removal** — and draws a firm boundary around downstream effectors (HINT1, transcription factors), the related Ap3A/Fhit system, and extracellular receptor-mediated ApnA signalling, all of which are adjacent systems rather than parts of the module.

The central architectural insight is an **asymmetry between the two halves of the module**. Synthesis is ancient and essentially universal: Ap4A is produced as a side reaction of aminoacyl-tRNA synthetases (aaRS), enzymes that are among the most deeply conserved proteins in all three domains of life. Two mechanistically distinct synthetic routes exist — the aminoacyl-adenylate ("LysRS type") route, in which the enzyme's own amino-acyl-adenylate intermediate donates AMP to the γ -phosphate of ATP, and the direct ATP-condensation ("GlyRS type") route, in which two ATP molecules are joined independently of the cognate amino acid. Removal, by contrast, has been **repeatedly reinvented**: at least three chemically distinct enzyme families clear Ap4A in different lineages — asymmetrical Nudix hydrolases (ATP + AMP; animals and most eukaryotes), symmetrical ApaH-type metallophosphatases (two ADP; proteobacteria, with an unrelated HD-domain YqeK family doing the equivalent job in Firmicutes), and fungal HIT-superfamily Ap4A phosphorylases (ADP + ATP by phosphorolysis). This is a textbook case of **non-orthologous gene displacement** layered on top of a conserved synthetic core.

Two live controversies frame the biology. First, whether Ap4A is a genuine regulatory signal or merely an unavoidable **damage/by-product metabolite** that cells constitutively dispose of — a question whose answer appears to depend on organism and concentration. Second, a reconciling discovery that Ap4A/Np4A also serve as **non-canonical 5' RNA caps**, which recasts several "removal" enzymes (ApaH, YqeK, NUDT2) as **decapping enzymes** that tune mRNA

stability, connecting the turnover module to the epitranscriptome. These findings do not settle the alarmone debate but reframe it: the molecule may be simultaneously a by-product, a cap donor, and — in specific eukaryotic contexts — a signal.

2. Definition and Biological Boundaries

What is inside the module

The Ap4A turnover module comprises exactly two reactions and the enzymes that catalyse them:

1. **Ap4A synthesis** by aminoacyl-tRNA synthetases (a side reaction of aminoacylation).
2. **Ap4A removal** by any one of three unrelated enzyme chemistries.

The relationship between the two halves is a simple supply-and-demand coupling: **aaRS supply the Ap4A that removal enzymes degrade**. The steady-state concentration of Ap4A in any given cell is therefore the ratio of these opposing fluxes, and because the removal enzyme differs by lineage, the "set point" of the module is tuned by non-homologous machinery in different organisms.

What is adjacent and should be treated separately

Several systems share molecular components, substrates, or vocabulary with Ap4A turnover but lie outside the module boundary:

- **The Ap3A / Fhit system.** Fhit (fragile histidine triad) is a HIT-superfamily dinucleoside-triphosphate hydrolase whose preferred substrate is Ap3A and which acts as a tumour suppressor. Critically, its tumour-suppressive function is *largely independent of its catalytic (Ap3A-hydrolase) activity* ([PMID: 41895442](#), [PMID: 23947369](#), [PMID: 25098403](#)). Fhit is a distinct metabolite (Ap3A) handled by a distinct protein for a distinct physiological output.
- **The HINT1 effector.** HINT1 is a separate HIT-family protein that *binds* aminoacyl-adenylates and Ap4A and acts as a transcriptional co-repressor in the LysRS–Ap4A–HINT1–MITF axis ([PMID: 22329685](#), [PMID: 19524539](#)). HINT1 is a **downstream reader** of Ap4A, not a turnover enzyme; it neither makes nor degrades the molecule.

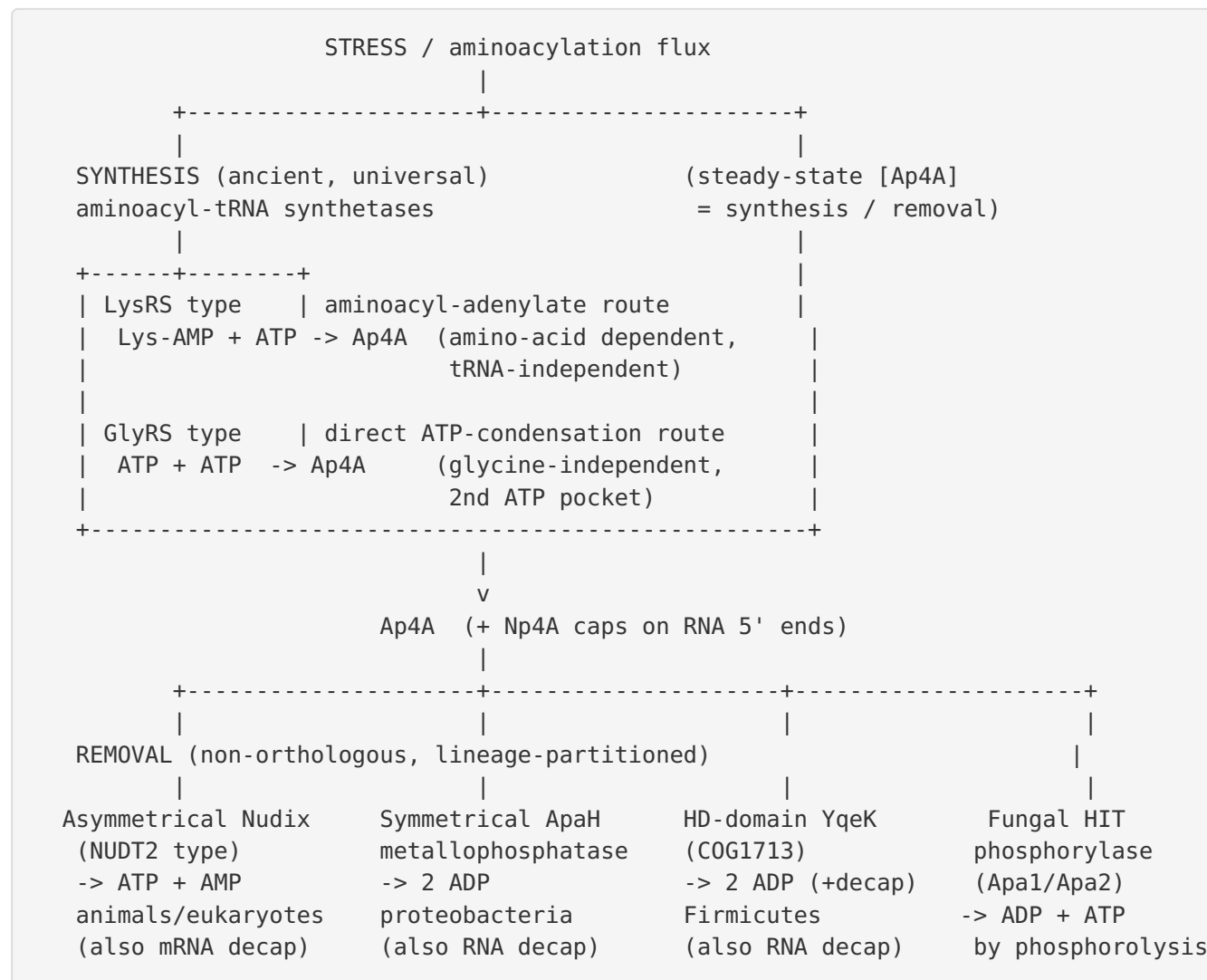
- **Extracellular ApnA purinergic signalling.** Diadenosine polyphosphates (Ap3A–Ap6A) are stored at high concentration in platelet dense granules in a metabolically inert form and released into the circulation on platelet activation, where they exert cardiovascular effects through P1/P2 purinoceptors and a proposed dedicated Ap4A receptor on ventricular myocytes ([PMID: 10434992](#), [PMID: 15320695](#)). This is a **receptor-mediated, extracellular** signalling system, mechanistically unrelated to intracellular synthesis/removal.

Competing definitions

The chief definitional fault line in the literature is "**alarmone/second messenger**" vs. "**damage metabolite**." This is not merely semantic: it determines whether the removal enzymes are regarded as *signal-terminating* (like a phosphodiesterase ending a cAMP signal) or as *housekeeping/detoxifying* enzymes (like other Nudix "house-cleaning" hydrolases). We treat both framings as viable and organism-dependent (see Section 7).

3. Mechanistic Overview

The best current model



Obligatory, conditional and accessory steps

- **Obligatory.** Ap4A synthesis is an obligatory *consequence* of aminoacylation chemistry — the aminoacyl-adenylate intermediate can be intercepted by ATP whenever tRNA is limiting, so some Ap4A production is essentially unavoidable wherever aaRS operate. Removal is obligatory in the sense that every cell examined possesses a dedicated hydrolase/phosphorylase; cells that cannot clear Ap4A accumulate it to toxic or phenotype-altering levels.

- **Conditional.** Which *synthetic route* dominates is conditional on the enzyme and the metabolic state: the LysRS route is amino-acid dependent and is stimulated under immune activation and antibiotic stress, whereas the GlyRS route is amino-acid *independent* and set by ATP concentration. Which *removal chemistry* operates is conditional on lineage (see Section 5).
- **Accessory.** The RNA-capping/decapping functions are accessory elaborations: Np4A caps are made when Np4A pools rise and removed by the same enzymes that clear free Ap4A, but capping is not required for the core turnover cycle.

4. Major Molecular Players and Active Assemblies

4.1 Synthesis — two class II aaRS mechanisms

Finding F001. Ap4A is synthesised by aminoacyl-tRNA synthetases via two mechanistically distinct routes, both using class II synthetases:

- **LysRS (aminoacyl-adenylate / "LysRS type") route.** Lysyl-tRNA synthetase first forms its lysyl-adenylate (Lys-AMP) intermediate, then transfers the AMP moiety to the γ -phosphate of a second ATP, yielding Ap4A. This reaction is amino-acid-dependent and proceeds **in the absence of tRNA** ([PMID: 26048731](#), [PMID: 19524539](#)). Mechanistically, "lysyl-tRNA synthetases efficiently produce diadenosine tetraphosphate (Ap4A) from lysyl-AMP with ATP in the absence of tRNA."
- **GlyRS (direct condensation / "GlyRS type") route.** Human glycyl-tRNA synthetase forms Ap4A by **direct condensation of two ATP molecules**, independent of glycine concentration, using a second ATP-binding pocket created by a GlyRS-specific insertion domain ([PMID: 19710017](#)): "human glycyl-tRNA synthetase (GlyRS) produces Ap4A by direct condensation of two ATPs, independent of glycine concentration."

The GlyRS insertion domain (ID1) that supports Ap4A synthesis is a rubredoxin-like zinc ribbon; mutations in this region are implicated in distal hereditary motor neuropathy and Charcot-Marie-Tooth disease, hinting at a possible link between Ap4A synthesis and neuronal function ([PMID: 25721219](#)). LysRS-type synthesis is broadly distributed — demonstrated for bacterial (*E. coli* LysU, *Myxococcus xanthus* LysS) and parasite (*Plasmodium falciparum*) enzymes ([PMID: 24736113](#), [PMID: 27392456](#), [PMID: 23633587](#)).

A notable **regulatory assembly** controls the LysRS route in mammals. In resting cells, LysRS is sequestered in the cytoplasmic **multi-tRNA-synthetase complex (MSC)**, bound to the scaffold protein p38/AIMP2. Phosphorylation of LysRS at Ser207 triggers a conformational change that disrupts its MSC-binding grooves, releasing LysRS, driving its nuclear translocation, and switching it from a translational to a transcriptional/Ap4A-producing mode ([PMID: 23159739](#)).

4.2 Removal — three chemistries, four families

Finding F002. Ap4A is removed by three unrelated enzyme chemistries partitioned by lineage. Chemically, the key distinction is *where* the Ap4A bridge is cleaved and whether the attacking nucleophile is water (hydrolysis) or phosphate (phosphorolysis):

Removal chemistry	Family / fold	Products	Lineage	Representative	Citation
Asymmetrical hydrolysis	Nudix / MutT superfamily	ATP + AMP	Animals, most eukaryotes	NUDT2 (human), CT771/nudH (<i>Chlamydia</i>), <i>C. elegans</i>	PMID: 24354275
Symmetrical hydrolysis	ApaH-type metallophosphatase	2 ADP	Proteobacteria	ApaH (<i>E. coli</i>)	PMID: 28516732
Symmetrical hydrolysis	HD-domain / COG1713	2 ADP	Firmicutes	YqeK (<i>B. subtilis</i>)	PMID: 32152217 , PMID: 41873760
Phosphorolysis	HIT / GalT superfamily	ADP + ATP	Fungi	Apa1 / Apa2 (<i>S. cerevisiae</i>)	PMID: 23628156 , PMID: 2174863

The asymmetrical Nudix hydrolases "asymmetrically cleave the metabolite Ap4A into ATP and AMP while facilitating homeostasis" ([PMID: 24354275](#)). The fungal route is distinctive in using **phosphorolysis** rather than hydrolysis: "the homeostasis of intracellular Ap4A in the yeast *Saccharomyces cerevisiae* is maintained by two 60% sequence-identical paralogs of Ap4A phosphorylases (Apa1 and Apa2)" ([PMID: 23628156](#)). The paralogs act in catabolism; an *apa1* *apa2* double mutant accumulates all bis(5'-nucleosidyl) tetraphosphates 5–50-fold yet remains

viable, showing the module is homeostatically important but not individually essential in yeast (PMID: 2174863). In Firmicutes, the HD-domain **YqeK** provides symmetrical activity structurally unrelated to ApaH and additionally "functions both as an Np4A hydrolase... and as a decapping enzyme" (PMID: 41873760).

4.3 The capping/decapping elaboration

Finding F004. Np4A/Ap4A act as non-canonical RNA 5' caps, linking the turnover module to the epitranscriptome, and the removal enzymes double as decapping enzymes. In *E. coli*, "RNAs acquire this epitranscriptomic modification during disulfide stress or when the dinucleoside tetraphosphate (Np4A) hydrolase ApaH is eliminated by mutation" (PMID: 41873760); ApaH both lowers the Np4A substrate pool and directly decaps Np4-capped RNA (PMID: 35131855, PMID: 40789943). The same logic operates in *B. subtilis* via YqeK, which converts Np4 caps to 5'-diphosphate ends to accelerate degradation (PMID: 41873760). In humans, NUDT2 is "a mRNA decapping and Ap4A hydrolysing enzyme" whose biallelic loss-of-function causes a neurodevelopmental disorder with a markedly altered transcriptome (PMID: 38141063).

5. Evolutionary and Cell-Biological Variation

5.1 Across evolutionary lineages

Finding F005. The module shows an ancient, universally distributed synthetic step combined with lineage-specific non-orthologous replacement of the removal step. Ap4A is "a side product of aminoacyl-tRNA synthetases found in all domains of life" (PMID: 28516732); because aaRS are among the most ancient, universally conserved enzymes, the capacity to make Ap4A is plausibly ancestral and effectively unavoidable. By contrast, the removal enzymes belong to several independent, structurally unrelated superfamilies:

- **Nudix / MutT-related "house-cleaning" hydrolases** — one of at least four house-cleaning NTPase superfamilies (PMID: 16359314): "House-cleaning NTP pyrophosphatases targeting non-canonical NTPs belong to at least four structural superfamilies: MutT-related (Nudix) hydrolases, dUTPase, ITPase (Maf/HAM1) and all-alpha NTP pyrophosphatases (MazG)."
- **HIT / GalT superfamily** — fungal Apa1/Apa2 phosphorylases (PMID: 23628156).
- **ApaH-type metallophosphatases** and the unrelated **HD-domain YqeK/COG1713** family (PMID: 32152217, PMID: 41873760).

The evolutionary pattern is therefore **a conserved core with a replaceable periphery**: the problem of clearing Ap4A has been solved independently at least four times by chemically different folds. This is the signature of **non-orthologous gene displacement**. For understanding the *ancestral* removal role, the best representatives are the broadly distributed house-cleaning Nudix hydrolases, which fit a detoxification origin; the fungal phosphorylase and the two bacterial symmetrical families are better understood as lineage-specific solutions, some of which (YqeK, ApaH) acquired RNA-decapping roles.

5.2 Across cell types, states and compartments

- **Immune activation (mammals).** In immunologically activated mast cells, Ser207-phosphorylated LysRS exits the MSC, enters the nucleus, and produces Ap4A, which binds the HINT1 repressor to release MITF and activate transcription ([PMID: 19524539](#), [PMID: 23159739](#)). This is a state- and compartment-specific (nuclear) deployment of the synthetic half of the module.
- **Innate immunity / STING.** LysRS-dependent Ap4A production has also been reported to curb STING-dependent inflammation, adding a second immune context in which the synthetic step is physiologically engaged ([PMID: 32494729](#)).
- **Antibiotic/oxidative stress (bacteria).** Aminoglycoside antibiotics elevate Ap4A, which enhances their bactericidal activity ([PMID: 31004054](#)); in *E. coli*, elevated Ap4A (via *apaH* deletion) modulates quorum sensing, biofilm formation and motility ([PMID: 37978430](#)). Disulfide (oxidative) stress is a principal trigger for Np4A-cap accumulation ([PMID: 35131855](#), [PMID: 41873760](#)).
- **Essentiality varies.** In the malaria parasite *Plasmodium berghei*, the Ap4A hydrolase is essential (an *ap4ah* null is non-viable), unlike several other house-cleaning enzymes ([PMID: 30700216](#)). In yeast, by contrast, the double phosphorylase mutant is viable ([PMID: 2174863](#)). Essentiality of the removal step is therefore organism-specific.

5.3 Alternative routes to the same outcome

The module is a striking example of **convergent problem-solving**. The same net task — keeping Ap4A low — is achieved by (i) hydrolysing to ATP + AMP, (ii) hydrolysing to 2 ADP, or (iii) phosphorylating to ADP + ATP. The products differ in their adenylate-energy content and in how they re-enter nucleotide metabolism, which may itself be adaptive in different metabolic contexts.

6. Constraints, Dependencies, and Failure Modes

Ordering and dependency constraints

- **Synthesis precedes and supplies removal.** The two halves are coupled only through the shared Ap4A pool; there is no obligate physical complex between synthetic and removal enzymes. Steady-state Ap4A is set by the *ratio* of the two fluxes.
- **The LysRS route requires a prior aminoacyl-adenylate.** The aminoacyl-adenylate must form (amino-acid-dependent) before AMP can be transferred to ATP; this route cannot proceed without the cognate amino acid. The GlyRS route has no such dependency (glycine-independent), so these two synthetic routes are **mechanistically mutually exclusive within a single catalytic cycle** even though both yield Ap4A.
- **tRNA competes with Ap4A synthesis.** Because tRNA is the physiological acceptor of the aminoacyl-adenylate, high tRNA availability channels flux toward aminoacylation and away from Ap4A; Ap4A synthesis is favoured when tRNA is limiting. (In *M. xanthus* LysS, tRNA^{Lys} inhibits Ap4A formation.)

Failure modes

- **Loss of removal → toxic/aberrant accumulation.** NUDT2 knockout raises intracellular Ap4A ~175-fold and remodels the transcriptome, down-regulating interferon/inflammation and tumour-promotion genes ([PMID: 27144453](#)). Biallelic NUDT2 loss-of-function causes a human neurodevelopmental disease ([PMID: 38141063](#)). In *E. coli*, only very high Ap4A is toxic, acting through zinc homeostasis and spurious binding at ATP sites ([PMID: 28516732](#)).
- **Dysregulated capping.** When ApaH/YqeK are lost, Np4-capped RNAs accumulate; controlled capping can be protective (elevated capping improves survival under disulfide stress) but unchecked capping dysregulates mRNA stability ([PMID: 41873760](#), [PMID: 40789943](#)).

Evidence that rules out otherwise-plausible paths

- Ap4A phosphorylase mutants show the fungal enzymes act in **catabolism, not synthesis**, in vivo (metabolite accumulation on knockout), ruling out a primarily anabolic role for Apa1/Apa2 ([PMID: 2174863](#)).

- In *E. coli*, the absence of any detectable Ap4A-triggered signalling cascade, together with constitutive non-regulated removal, argues against a dedicated signalling pathway in that organism ([PMID: 28516732](#)).

7. Controversies and Open Questions

7.1 Alarmone vs. damage metabolite

Finding F003. Ap4A's status as a signalling alarmone versus a damage/by-product metabolite is unresolved, and the best current reading is that the answer is **organism- and concentration-dependent**.

- **Pro-signalling (eukaryotes).** The LysRS–Ap4A–HINT1–MITF axis is a concrete, structurally characterised signalling pathway: "silencing LysRS led to reduced Ap4A production in immunologically activated cells, which resulted in a lower level of MITF inducible genes" ([PMID: 19524539](#)). NUDT2 knockout's 175-fold Ap4A rise and coordinated transcriptome remodelling are consistent with a regulatory role ([PMID: 27144453](#)).
- **Damage-metabolite null hypothesis (bacteria).** A systematic *E. coli* study found "no indications for a signaling cascade that is triggered by Ap4A... rather, we found that Ap4A is efficiently removed in a constitutive, nonregulated manner," and Ap4A met the formal criteria for a damage metabolite ([PMID: 28516732](#)).

A key caution is **cross-organism extrapolation**: the signalling evidence is strongest in mammalian immune cells, while the damage-metabolite evidence is strongest in *E. coli*. These may both be correct for their respective systems.

7.2 The reconciling RNA-cap hypothesis

The discovery that Np4A are 5'-RNA caps offers a mechanistic reconciliation: Ap4A can be an unavoidable by-product, a cap-donor substrate, and (in some lineages) a signal all at once, with the "removal" enzymes doubling as decapping enzymes that tune the epitranscriptome ([PMID: 35131855](#), [PMID: 40789943](#), [PMID: 41873760](#), [PMID: 38141063](#)). Whether capping is the *primary* biological rationale for maintaining low free Ap4A, or an opportunistic secondary use, is an open question.

7.3 Most important open questions

1. **Is there a conserved eukaryotic Ap4A signalling pathway, or is the LysRS–HINT1–MITF axis cell-type-restricted?**
 2. **What fraction of Ap4A flux is dedicated to RNA capping vs. free-pool homeostasis, and does this differ between bacteria, fungi, and animals?**
 3. **Why have lineages repeatedly replaced the removal enzyme** while conserving the synthetic side — is product identity (ATP+AMP vs. 2 ADP vs. ADP+ATP) under selection?
 4. **Are the mechanistically distinct synthetic routes (LysRS vs. GlyRS) functionally specialised** for different physiological triggers?
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8. Mechanistic Model / Interpretation (Synthesis)

The Ap4A turnover module is best understood as **a conserved generator coupled to an interchangeable eraser**. The generator — aminoacyl-tRNA synthetase side chemistry — is an inevitable consequence of how aaRS activate amino acids, and it is as old as the genetic code itself. Because it is unavoidable, every cell faces the same downstream problem: a dinucleoside polyphosphate that rises under stress and can perturb nucleotide-sensing machinery if allowed to accumulate. Evolution has solved that problem **repeatedly and independently**, recruiting whichever hydrolase or phosphorylase fold was locally available (Nudix, ApaH metallophosphatase, HD-domain, or HIT/GalT). The result is a module whose two halves have wildly different evolutionary tempos — a slow, conserved input and a fast, convergently reinvented output.

The alarmone-vs-damage debate is best read not as a contradiction but as a reflection of this architecture. A molecule that is produced unavoidably and cleared constitutively looks exactly like a damage metabolite — and in a plain bacterium it may be nothing more. But the same molecule, in a cell that has evolved a reader (HINT1) and a regulated, compartment-switching producer (phospho-LysRS leaving the MSC for the nucleus), becomes a bona fide signal. The RNA-capping discovery adds a third layer: the molecule is also a covalent cap donor, so its clearance enzymes are simultaneously decapping enzymes. The module is therefore **one chemistry wearing three hats** — by-product, cap, and signal — with the balance among them set by lineage and physiological state.

9. Evidence Base

PMID	How it supports/challenges the findings
19710017	Establishes the GlyRS direct-ATP-condensation synthesis route (F001).
26048731	Establishes the LysRS aminoacyl-adenylate synthesis route, tRNA-independent (F001).
23159739	Structural basis for LysRS release from the MSC and translation→transcription switch (F001/F003).
24354275	Defines asymmetrical Nudix (NUDT2-type) removal → ATP + AMP (F002).
23628156	Fungal Apa1/Apa2 HIT-family phosphorylase homeostasis in yeast (F002/F005).
2174863	Shows Apa1/Apa2 act in catabolism; double mutant viable, accumulates ApnA (F002).
32152217	Identifies YqeK/COG1713 as a novel Ap4A hydrolase family (F002/F005).
41873760	YqeK as both Np4A hydrolase and decapping enzyme in Firmicutes; ApaH controls caps (F002/F004).
28516732	Damage-metabolite null hypothesis in <i>E. coli</i> ; universal aaRS origin (F003/F005).
19524539	LysRS/HINT1/MITF signalling axis in immune cells (F003/F006).
27144453	NUDT2 KO raises Ap4A 175-fold and remodels transcriptome (F003).
35131855	Np4A as RNA-cap precursors; RNA-recognition mechanism (F004).
40789943	ApaH directly decaps Np4-capped RNA (F004).
38141063	Human NUDT2 is an mRNA decapping enzyme; biallelic loss causes disease (F004).
16359314	Nudix among four ancient house-cleaning NTPase superfamilies (F005).
41895442	Fhit tumour suppression independent of Ap3A-hydrolase activity — adjacent system (F006).
22329685	HINT1 as a downstream Ap4A-binding effector, outside the module (F006).
10434992	Extracellular ApnA storage/release from platelets — separate system (F007).
15320695	Extracellular ApnA act via purinoceptors/dedicated Ap4A receptor (F007).
31004054	Ap4A elevated by aminoglycosides, enhances bactericidal activity (variation).
30700216	Ap4A hydrolase essential in <i>Plasmodium berghei</i> (variation).
32494729	LysRS-produced Ap4A curbs STING-dependent inflammation (variation).

PMID	How it supports/challenges the findings
25721219	GlyRS ID1 zinc-ribbon; disease mutations link to Ap4A synthesis (synthesis detail).

10. Limitations and Knowledge Gaps

- **Organism-mixing.** The strongest signalling data come from mammalian immune cells, the strongest "damage" data from *E. coli*, and the capping data from bacteria plus one human disease gene. Conclusions should not be generalised across these systems without caution.
- **Concentration realism.** Many phenotypes derive from genetic knockouts that produce supraphysiological Ap4A (e.g. 175-fold in NUDT2-KO). Whether endogenous fluctuations reach signalling thresholds in normal physiology is often unestablished.
- **Flux partitioning unknown.** The fraction of Ap4A committed to RNA capping vs. free-pool homeostasis has not been quantified in any system.
- **Synthetic-route specialisation untested.** Whether LysRS-type and GlyRS-type synthesis serve distinct physiological triggers remains inferential.
- **This review is literature-based.** No primary datasets were analysed; findings rest on the cited experimental literature.

11. Proposed Follow-up Experiments / Actions

1. **Quantify Ap4A flux partitioning.** Use metabolic labelling plus Np4A-cap sequencing in a single organism (e.g. *E. coli* and a yeast) to measure what fraction of Ap4A is capped onto RNA vs. hydrolysed in the free pool.
2. **Test synthetic-route specialisation.** Compare LysRS- vs. GlyRS-dependent Ap4A production under matched stresses (amino-acid starvation, oxidative stress, antibiotic challenge) using route-specific catalytic mutants.
3. **Dose-resolved signalling thresholds.** Titrate intracellular Ap4A (e.g. inducible NUDT2 degron) to determine the concentration at which transcriptome remodelling begins, distinguishing signalling from toxicity.

4. **Decapping-vs-hydrolysis separation of function.** Engineer separation-of-function alleles of ApaH/YqeK/NUDT2 that cleave free Ap4A but not caps (or vice versa) to assign phenotypes to each activity.
5. **Cross-lineage complementation.** Test whether a fungal phosphorylase, a Nudix hydrolase, and an ApaH metallophosphatase are interchangeable in restoring Ap4A homeostasis in a removal-deficient host, probing whether product identity (ATP+AMP vs 2 ADP vs ADP+ATP) matters physiologically.
6. **Phylogenomic census.** Systematically map which removal family each sequenced lineage uses to refine the non-orthologous-displacement picture and identify lineages with losses or redundancy.

Consensus answer

Ap4A turnover is a reusable two-part module: an ancient, universal synthetic activity (Ap4A made as an aminoacyl-tRNA-synthetase side reaction, by LysRS-type transfer of an aminoacyl-adenylate AMP to ATP or by GlyRS-type direct condensation of two ATPs) balanced against a removal step that has been repeatedly replaced by non-homologous enzymes — asymmetric Nudix hydrolases (ATP+AMP; animals/most eukaryotes), HIT-family phosphorylases (ADP+ATP; fungi), and symmetric metallophosphatases (2 ADP; ApaH in proteobacteria, HD-domain YqeK in Firmicutes). Whether Ap4A is a genuine alarmone or an unavoidable damage by-product is organism- and concentration-dependent; a reconciling view is that Ap4A/Np4A also serve as non-canonical RNA 5' caps, so the removal enzymes double as decapping enzymes. The adjacent Ap3A/Fhit system, the HINT1 effector, and extracellular ApnA purinergic signalling share components or vocabulary but lie outside the module.

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